Algebraic Coding Theory

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Abstract

Notes taken while studying Algebraic Coding Theory, mostly from Hill book [1] and Lint book [2].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to $LaTeX$.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

Contents

1 Vector spaces over finite fields, quick recap

Let q be a prime power.

$V(n,q)$: set $GF(q)^n$ of all ordered n-tuples over $GF(q)$. $V(n,q)$ is a vector space.

A set of vectors $\{v_1, v_2, \ldots, v_r\}$ is said to be *linear dependent* if $\exists$ scalars $a_1, \ldots, a_r$, not all zero, such that

$$a_1v_1 + a_2v_2 + \ldots + a_rv_r = 0$$

They are *linear independent* if

$$a_1v_1 + a_2v_2 + \ldots + a_rv_r = 0 \overset{implies}{\Longrightarrow} a_1 = a_2 = \ldots = a_r = 0$$

Let C a subspace of $V(n,q)$. Then a subset $\{v_1, v_2, \ldots, v_r\} \subseteq C$ is called a *generating set* (or *spanning set*) of C if every vector in C can be expressed as a linear combination of $v_1, \ldots, v_r$.

A generating set of C which is also linearly independent is called a *basis* of C.

Theorem 4.3. Suppose $\{v_1, v_2, \ldots, v_k\}$ is a basis of a subspace C of $V(n,q)$. Then

i. every vector of C can be expressed uniquely as a linear combination of the basis vectors.

ii. C contains exactly q^k vectors.

Proof. i. suppose vector $x \in C$ such that

$$x = a_1v_1 + a_2v_2 + \cdots + a_kv_k$$

$$x = b_1v_1 + b_2v_2 + \cdots + b_kv_k.$$

then

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \cdots + (a_k - b_k)v_k = 0.$$

But the set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent, and so

$$a_i - b_i = 0 \quad \text{for } i = 1, 2, \ldots, k,$$

i.e.,

$$a_i = b_i \quad \text{for } i = 1, 2, \ldots, k.$$

ii. By (i), the q^k vectors

$$\sum_{i=1}^{k} a_iv_i \qquad (a_i \in \mathrm{GF}(q))$$

are precisely the distinct vectors of C.

□

2 Linear codes

The theorems of this section are from [1] and follow the original enumeration of the book.

Note about notation:
A q-ary linear code $[n,k,d]$-code is also a q-ary (n, q^k, d)-code, but the opposite is not always true.

2.1 Some definitions

Let q be a prime power. View $(F_q)^n$ as the vector space $V(n, q)$.

Definition (Linear code over GF(q))**.** A *linear code over* $GF(q)$ is a subspace of $V(n, q)$, with $n \geq 0$.

Thus $C \subseteq V(n, q)$ is a linear code iff

i. $u + v \in C \quad \forall u, v \in C$

ii. $au \in C \quad \forall u \in C,\ a \in GF(q)$

If C is a k-dimensional subspace of $V(n, q)$, then the linear code C is called an $[n, k]$-code. (sometimes with minimum distance d: $[n, k, d]$-code).

Definition (Weight)**.** Weight, $w(x)$, of a vector $x \in V(n, q)$ is the number of non-zero entries of x.

Lemma 5.1. if $x, y \in V(n, q)$, then $d(x, y) = w(x - y)$.

Proof. the vector $x - y$ has non-zero entries in preceisely those places where x and y differ. □

Theorem 5.2. let C a linear code, let $w(C)$ be the smallest of weights of the non-zero codewords of C. Then $d(C) = w(C)$.

Proof. $\exists\ x, y \in C$ s.th. $d(C) = d(x, y)$.
By Lemma 5.1, $d(C) = w(x - y) \geq w(C)$, since $x - y$ is a codeword of the linear code C (ie. $x - y \in C$).
On the other hand, for $x \in C$,
$w(C) = w(x) = d(x, 0) \geq d(C)$, since $0 \in C$.
Hence, $d(C) \geq w(C)$ and $w(C) \geq d(C) \implies d(C) = w(C)$. □

Definition (Generator matrix (GM))**.** a $k \times n$ matrix whose rows form a basis of a linear $[n, k]$-code is called a *generator matrix* (GM) of the code.

2.2 Equivalence

Theorem 5.4. two known $k \times n$ matrices generate equivalent linear $[n, k]$-codes over $GF(q)$ if one matrix can be obtained from the other by a sequence of operations of the following types:

R1. permutation of the rows

R2. multiplication of a row by a non-zero scalar

R3. addition of a scalar multiple of one row to another

C1. permutation of the columns

C2. multiplication of any column by a non-zero scalar

Theorem 5.5. let G a GM of an $[n, k]$-code. Then by performing the previous operations, G can be transformed to the standard form $[I_k | A]$ where I_k is the $k \times k$ identity matrix, and A is a $k \times (n - k)$ matrix.

2.3 Encoding with a linear code

Let C a $[n,k]$-code over $GF(q)$ with GM G. C contains q^k codewords $\Longrightarrow$ can be used to communicate any one of q^k distinct messages.

Iddentify these messages with the q^k tuples of $V(k,q)$.

Encoding:

msg vector: $u = u_1u_2 \dots u_k$; rows of G: $r_1r_2 \dots r_k$. Encode

$$uG = \sum_{i=1}^{k} u_i r_i \in C$$

ie. uG is a codeword of C; a linear combination of the rows of the GM.

Encoding function:

$$\begin{aligned} V(k,q) &\longrightarrow C \subseteq V(n,q) \\ u &\longmapsto uG \end{aligned}$$

If G is in standard form (5.5), $G = [I_k|A]$ with $A = [a_{ij}]$, then encoding u is:

$$x = uG = x_1x_2 \dots x_k x_{k+1} \dots x_n$$

where

$$\begin{aligned} x_i &= u_i \quad \text{for } 1 \leq i \leq k, \text{ message digits} \\ x_{k+i} &= \sum_{j=1}^{k} a_j i u_j \quad \text{for } 1 \leq i \leq n-k, \text{ check digits} \end{aligned}$$

the check digits add redundancy to gie protection against noise.

2.4 Decoding with a linear code

Codeword $x = x_1x_2 \dots x_n$, received vector $y = y_1y_2 \dots y_n$, error vector $e = y - x = e_1e_1 \dots e_n$.

2.4.1 Cosets

Cosets here are similar to the ones from typical group theory, just from another perspective.

Definition (Coset)**.** Let C a $[n,k]$-code over $GF(q)$ and $a \in V(n,q)$. Then the set $a + C = \{a + x | x \in C\}$ is called a *coset* of C.

Definition (Coset leader)**.** the vector having minimum weight in a coset is called the *coset leader*.

Lemma 6.3. Let $a + C$ a coset of C and $b \in a + C$. Then $b + C = a + C$.

Proof. since $b \in a + C \implies b = a + x$ for some $x \in C$. If $b + y \in b + C$ then

$$b + y = (a + x) + y = a + (x + y) \in a + C$$

For the other direction, if $a + x \in a + C$ then

$$a + x = (b - x) + z = b + (z - x) \in b + C$$

hence $a + C \subseteq b + C$.

Therefore $b + C = a + C$. $\square$

Theorem 6.4 (Lagrange)**.** suppose $C \subseteq V(n,q)$ is a $[n,k]$-code over $GF(q)$. Then

i. evvery vector of $V(n,q)$ is in some coset of C

ii. every coset contains exactly q^k vectors

iii. two cosets either are disjoint or coincide (partial overlap is impossible)

Proof. i. if $a \in V(n,q)$ then $a = a + 0 \in a + C$

ii. the map

$$\begin{aligned} C &\longrightarrow a + C \ \forall \ x \in C \\ x &\longmapsto a + x \end{aligned}$$

is a one-to-one map, hence $|a + C| = |C| = q^k$.

iii. by contradiction, suppose cosets $a+C$, $b+C$ overlap. Then for some vector v,

$$v \in (a + C) \cap (b + C)$$

thus for some $x, y \in C$, $v = a + x = b + y$.

Hence $b = a + (x - y) \in a + C$, thus by Lemma 6.3 we have $b + C = a + C$. $\square$

Corollary . 6.4 shows that $V(n,q)$ is partitioned into disjoint cosets of C:

$$V(n,q) = (0 + C) \cup (a_1 + C) \cup \ldots \cup (a_s + C)$$

where $s = q^{n-k} - 1$, and by Lemma 6.3 we may take $0, a_1, \ldots, a_s$ to be coset leaders.

Example 6.5. Let C be the binary $[4,2]$-code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

ie. $C = \{0000,\ 1011,\ 0101,\ 1110\}$.

Then the cosets of C are

$$\begin{aligned}
&0000 + C = C \text{ itself} \\
&1000 + C = \{1000,\ 0011,\ 1101,\ 0110\} \\
&0100 + C = \{0100,\ 1111,\ 0001,\ 1010\} = 0001 + C \\
&0010 + C = \{0010,\ 1001,\ 0111,\ 1100\}
\end{aligned}$$

by Lemma 6.3: $0001 \in 0100 + C$, thus $0001 + C = 0100 + C$.

2.5 Probability of error correction (in binary linear codes)

To evaluate how good a code is, must calculate the *probability* that a received vector will be decoded as the codeword which was sent.

Let p be the symbol error probability.

Probability that the error vector is agiven vector of weight i is $p^i(1-p)n-i$.

Theorem 6.7. Let C a binary $[n,k]$-code, and for $i = 0, 1, \ldots, n$ let α_i denote the number of coset leaders of weight i.

Then the probability $P_{corr}(C)$ that a received vector decoded by means of a standard array is the codeword which was sent is

$$P_{corr}(C) = \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i}$$

Example 6.8. $[n,k]$-code from 6.5, coset leaders are $0000, 1000, 0100, 0010$. Hence $\alpha_0 = 1,\ \alpha_1 = 3,\ \alpha_2 = \alpha_3 = \alpha_4 = 0$, so

$$P_{corr}(C) = (1-p)^4 + 3p(1-p)^3 = (1-p)^3(1+2p)$$

Lemma 6.9 (Remark 6.9)**.** if $d(C) = 2t+1$ or $d(C) = 2t+2$, then C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

Definition (Rate)**.** $[n,k]$-code C uses n symbols to send k message symbols; it has *rate* $R(C) = k/n$.

A good code will have a high rate.

Definition (Capacity)**.** the *capacity* $\mathscr{C}(p)$ of a binary symmetric channel with symbol error probability p is

$$\mathscr{C}(p) = 1 + p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p)$$

Theorem 6.12 (Shanon's theorem)**.** channel binary symmetric with symbol error probability p.

Let $R < \mathscr{C}(p)$.

Then for any $\epsilon > 0$, $\exists$ for sufficient large n, an $[n,k]$-code C of rate $k/n \geq R$ such that $P_{err}(C) < \epsilon$.

Definition (P_{symb})**.** P_{symb} denotes the average probability that a message symbol is in error after decoding.

Theorem 6.14. let C a binary [n,k]-code and let A_i denote the number of codewords of C of weight i.

Then, if C is used for error detection, the probability of an incorrect message being undetected is

$$P_{undetec}(C) = \sum_{i=1}^{n} A_i \cdot p^i(1-p)^{n-i}$$

Example 6.15. for the code of 6.5,

$$P_{undetec}(C) = \underbrace{1 \cdot p^2(1-p)^2}_{0101} + \underbrace{2 \cdot p^3(1-p)^{4-3=1}}_{1011,\ 1110} = p^2 - p^4$$

2.6 Exercises

Exercise 5.1. is the binary (11, 24, 5)-code linear?

Proof. recall: a q-ary linear $[n,k,d]$-code is also a q-ary (n, q^k, d)-code, but the opposite is not always true.

$\Longrightarrow$ $(11, 24, 5)$-code to be linear it would mean that $24 = q^k$.

Since it is binary, $q = 2$. But $24 \neq 2^k$, so it can not be linear. □

Exercise 5.2. E_n is the code of all even-weight vectors of $V(n,2)$, which is linear. What are the parameters $[n,k,d]$ of E_n? Write a GM for E_n in standard form.

Proof.

$$E_n = \{x \in V(n,2) | w(x) \text{ is even}\}$$

$\longrightarrow$ is the binary even-weight ode (the single parity-check code); it's linear since

- the zero vector has even weight
- over $GF(2)$, the sum of two even-weight vectors has even weight

Recall: $F_q^n \Longleftrightarrow V(n,q)$, so $V(n,2) \cong F_2^n$

$$\Longrightarrow \quad E_n = \{(x_1, \dots, x_n) \in F_2^n | x_1 + \ldots + x_n = 0\}$$

minimum distance is 2, since

- no word of weight 1 is in E_n
- words of weight 2 are in E_n

Therefore, parameters are $[n, n-1, 2]$
GM:

$$G = [I_{n-1}|1]$$

which generates vectors of the form

$$(a_1, \ldots, a_{n-1}, a_1 + \ldots + a_{n-1})$$

$\square$

Exercise 5.5. Prove that in a *binary linear code*, either all the codewords have even weight or exactly half have even weight and half odd weight.

Proof. let $C_e,\ C_o$ be subsets of C with even and odd weights respectively.
So that $C = C_e \cup C_o$ (disjoint union).
For binary vectors x, y:

$$w(x+y) = w(x) + w(y) \pmod 2$$

ie. over F_2, the parity of the weight is additive:

$$\begin{aligned} even + even &= even \\ even + odd &= odd \\ odd + odd &= even \end{aligned}$$

Now we have two possible cases:

- C has no odd weighted codewords:
  $C_0 = \emptyset,\ \forall x \in C\ x$ is even weighted
- C has at least one odd-weighted codeword:
  let $u \in C_o$, consider the map

  $$\begin{aligned} \phi : C_e &\longrightarrow C_o \\ c &\longmapsto c + u \end{aligned}$$

  with $c \in C_e,\ u \in C_o,\ \phi(c) \in C_o$.
  ϕ is bijective, since $\phi(\phi(c)) = (c+u) + u = c$,
  so ϕ is it's own inverse; hence $|C_e| = |C_o|$.
  Since $C = C_e \cup C_e$, half codewords are even weighted and half odd.

$\square$

Exercise 6.1. construct standard arrays for codes having each of the GMs:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Proof. recall: for a binary code with GM G, the code C consists of all binary linear combinations of the rows of G.

For G_1:
rows: $r_1 = (1,0),\ r_2 = (0,1))$. A codeword is any linear combination:

$$uG_1 = a \cdot r_1 + b \cdot r_2, \quad \text{with } a, b \in \{0,1\}$$

4 combinations:

$$\begin{aligned}
0 \cdot r_1 + 0 \cdot r_2 &= (0,0) \\
1 \cdot r_1 + 0 \cdot r_2 &= (1,0) \\
0 \cdot r_1 + 1 \cdot r_2 &= (0,1) \\
1 \cdot r_1 + 1 \cdot r_2 &= (1,0) + (0,1) = (1,1)
\end{aligned}$$

$$\implies C_1 = \{00, 10, 01, 11\}$$

$$\text{standard array: } 00\ 01\ 10\ 11$$

For G_2:
rows: $r_1 = (1,0,1),\ r_2 = (0,1,1)$. Codeword: $a \cdot r_1 + b \cdot r_2$ with $a, b \in \{0,1\}$.

4 combinations:

$$\begin{aligned}
0 \cdot r_1 + 0 \cdot r_2 &= (0,0,0) \\
1 \cdot r_1 + 0 \cdot r_2 &= (1,0,1) \\
0 \cdot r_1 + 1 \cdot r_2 &= (0,1,1) \\
1 \cdot r_1 + 1 \cdot r_2 &= (1,0,1) + (0,1,1) = (1,1,0)
\end{aligned}$$

$$\implies C_2 = \{000, 101, 011, 110\}$$

$\implies$ so the 1st row of the standard array is: 000 101 011 110.

Now, pick a non used vector: 001, add it to the code C_2:

$$\begin{aligned}
&001 + C_{2i} \\
&001 + 000 = 001 \\
&001 + 101 = 100 \\
&001 + 011 = 010 \\
&001 + 110 = 111
\end{aligned}$$

$\implies$ 2nd row of the standard array: 001 100 010 111

Thus the standard array is

$$\begin{aligned}
&000\ 101\ 011\ 110 \\
&001\ 100\ 010\ 111
\end{aligned}$$

ie. $2^3 = 8$ vectors, a $V(3,2)$ vector space. □

Exercise 6.4. C a binary $[n,k]$-code with minimum distance $2t+1$ (or $2t+2$). Given that p is very small, show that an approximate value of $P_{err}(C)$ is $\left(\binom{n}{t+1} - \alpha_{t+1}\right) p^{t+1}$, where α_{t+1} is the number of coset leaders of C of weight $t+1$.

Proof. From 6.9, if $d(C) = 2t+1$ (or $2t+2$) *Longrightarrow* C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

$$P_{corr}(C) = \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i} \tag{1}$$

where α_i is the number of coset leaders of weight i

From the $\binom{n}{i}$ vectors of weight i, exactly α_i are coset leaders, so the non coset leaders are $\binom{n}{i} - \alpha_i$. This is when a decoding error happens.

$$\Longrightarrow P_{err}(C) = \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i}$$

Alternatively from 1:

$$P_{err}(C) = 1 - P_{corr}(C) = 1 - \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i}$$

Notice that

$$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} = (p + (1-p))^n = 1^n = 1$$

thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} - \sum_{i=0}^{n} \alpha_i \cdot p^i (1-p)^{n-i} \\ &= \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \end{aligned}$$

Now, from 6.9, $\forall i \leq t,\ \alpha_i = \binom{n}{i}$, thus $\binom{n}{i} - \alpha_i = 0$ for the first t terms in the sum.

Thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^{n} \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \\ &= \left(\binom{n}{t+1} - \alpha_{t+1} \right) p^{t+1} (1-p)^{n-t+1} + \dots \end{aligned}$$

where the ... represent the terms with p^{t+2} and higher.

Then, since p is very small (by the exercise definition), $(1-p)^{n-t+1} \equiv 1$, and the other higher powers of p are negligible.

Hence,

$$P_{err}(C) = \left(\binom{n}{t+1} - \alpha_{t+1} \right) \cdot p^{t+1}$$

□

References

[1] Raymond Hill. A First Course in Coding Theory, 1986.

[2] J.H. van Lint. Introduction to Coding Theory, 1981.